

Description of possible interface with tpaCAD 4.0

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General

There are 4 different ways to convert external data so that they can be used on the FOMART4 CNC-machine.

- Use an external **post-prozessor** that generates *.tcn files (Tpacad file-format). These files can be used on the machine without additional manipulation.
- Use **2D or 3D DXF files**, if required with defined layer definition. These files can be opened in TpaCAD and be converted into *.tcn files
- Use a defined **iso-code format** to generate *.txt or *.iso files that can be opened in TpaCAD and also executed on the machine without additional manipulation.
- For **5-axis interpolation** movements *.txt or *.iso files with a defined file format are used. These files don't include information about tools and they have to be called using a subiso call in a normal *.tcn file. They can not be executed on the machine without additional programming. Offcourse it is possible to generate the *.tcn file including the subiso call, tool information and path+filename definition of the necessary *.txt or *.iso file via postprozessor and save the *.txt and *.iso file with the defined file name under the defined path. This function can also be used on 3-4 axis machines but offcourse without change of the BETA angle.

Post-prozessor Interface Specifications for external CAD

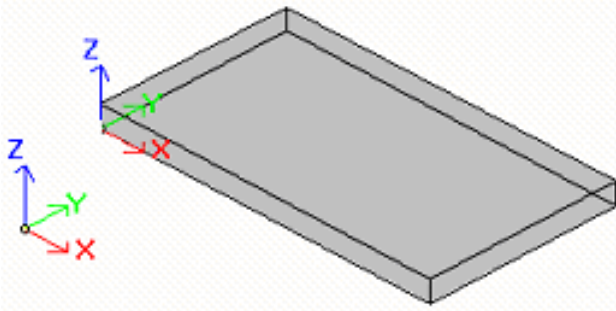
Description of programs format used by TpaCAD

Piece Geometry

Piece is a block object, featured by:

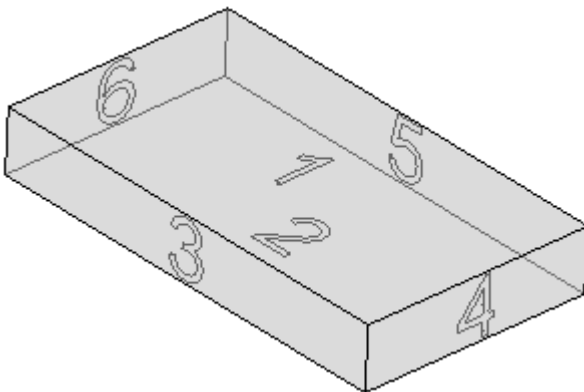
- Three dimensions: Length, height and thickness. Three piece dimensions are described by **l**, **h**, **s** letters.
- Six faces.

CAD use a three-dimensional system of fixed Cartesian coordinates, calls **Absolute Reference System of the piece (ARS)**, same for all pieces and placed like following figure show:



- Axis show: X, Y and Z
- System origin is placed in the left-below corner
- X axis is along piece length (described by: l) and have positive direction towards right.
- Y axis is along piece height (described by: h) and have positive direction towards piece inside.
- Z axis is along piece thickness (described by: s) and have positive direction towards up.

Six faces of the block call **Real Faces** and have numbers from 1 to 6. Following figure show the automatic enumeration of faces:



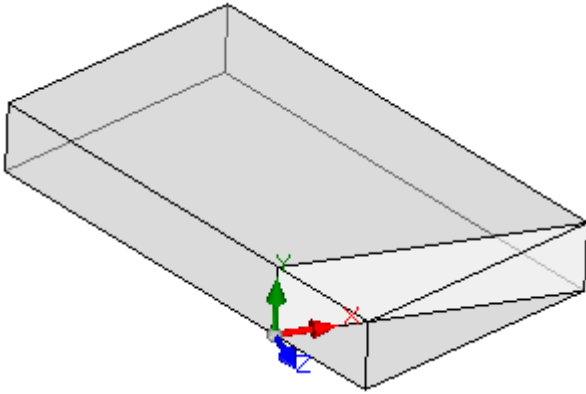
- upper face is the number 1
- below face is the number 2
- front face is the number 3
- right lateral face is the number 4
- back face is the number 5
- left lateral face is the number 6.

CAD can be configured to work with a different faces enumeration from the automatic one: in this case the enumeration called **custom**, and use numbers from 1 to 6 the same.

CAD can be configured to exclude (disable) one or more real faces.

The default execution sequence is: F1, F2, F3, F4, F5, F6

In addition to six real faces it's possible to create more face, usually placed inside the piece, call **fictitious faces**



Fictitious faces have numbers from 7 to 99 and can be:

- internal, external or through the piece
- whatever orientation respect the **ARS**

Programming workings in the piece is always relative to one face and referred to the three-dimensional Cartesian coordinates of the face. In this case there's a triad of axis XYZ, where:

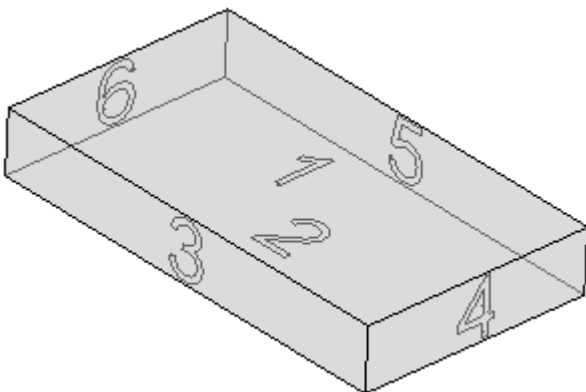
- face plane assigned axis X and Y
- the perpendicular direction of the face assigned axis Z, we use like depth axis.

In the **Face Reference System**:

- X axis is along the face length (described by: **lf**)
- Y axis is along the face height (described by: **hf**)
- Z axis is along the face thickness (described by: **sf**).

Following are described reference systems of real faces, in a standard configuration:

Faces 0:

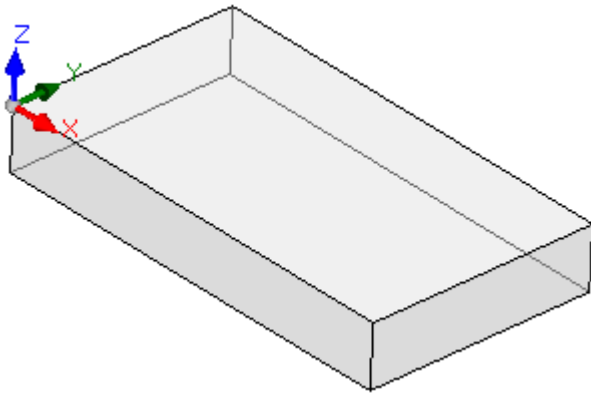


Face 0:

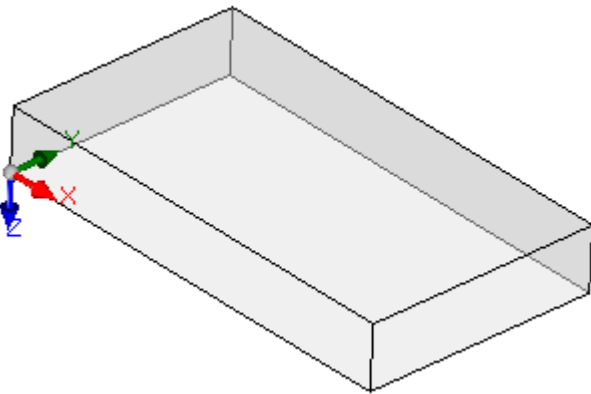
face dimensions and zero-points are always referred to the application face that is selected

This is a overall face that can be used for programming operations in all available faces. The execution sequence is only affected by the sequence of the lines and not the face sequence. In this way it is possible to sort operations that are on different faces.

Faces 1 and 2:

**Face 1:**

face dimensions

 $lf=l$ $hf=h$ $sf=s$ **Face 2:**

face dimensions:

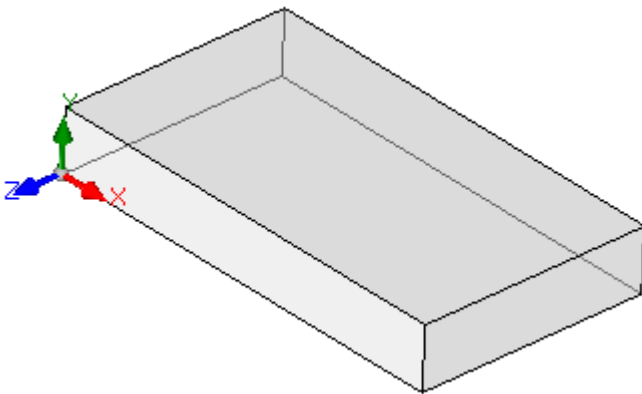
 $lf=l$ $hf=h$ $sf=s$

Local systems of faces are similar:

- X axis has same orientation and direction of X axis of **ARS**
- Y axis has same orientation and direction of Y axis of **ARS**
- Z axis has same orientation of Z axis of **ARS**, but in face 2 has reverse direction.

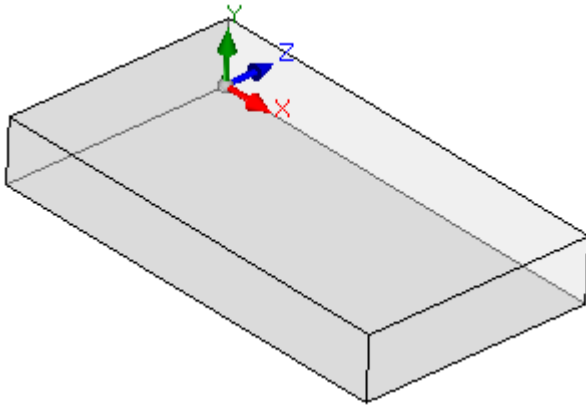
Compared to **ARS**, origin point of faces:

- in face 1 is placed at $(0; 0; s)$;
- in face 2 is placed at $(0; 0; 0)$.

Faces 3 and 5:**Face 3:**

face dimensions

 $lf=l$ $hf=s$ $sf=h$

**Face 5:**

face dimensions

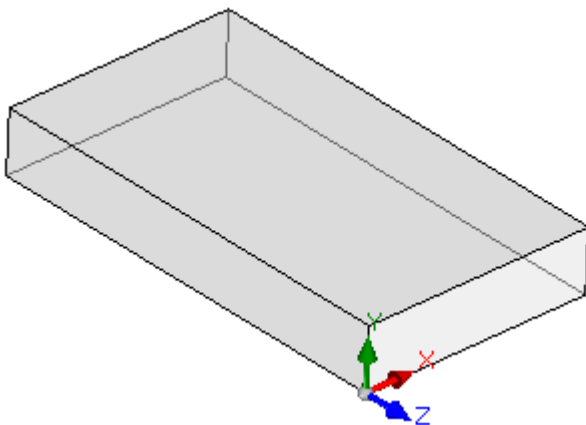
 $lf=l$ $hf=s$ $sf=h$

Local systems of faces are similar:

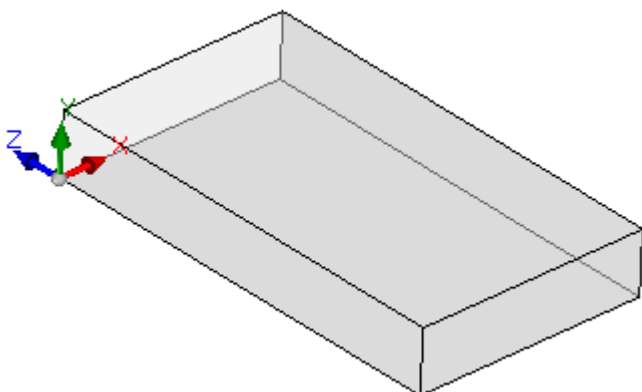
- X axis has same orientation and direction of X axis of **ARS**
- Y axis has same orientation and direction of Z axis of **ARS**
- Z axis has same orientation of Y axis of **ARS**, but in face 3 has reverse direction.

Compared to **ARS**, origin point of faces:

- in face 3 is placed at (0; 0; 0)
- in face 5 is placed at (0; h; 0).

Faces 4 and 6:**Face 4:**

face dimensions

 $lf=h$ $hf=s$ $sf=l$ **Face 6:**

face dimensions

 $lf=h$ $hf=s$ $sf=l$

Local systems of faces are similar:

- X axis has same orientation and direction of Y axis of **ARS**
- Y axis has same orientation and direction of Z axis of **ARS**
- Z axis has same orientation of X axis of **ARS**, but in face 6 has reverse direction.

Compared to **ARS**, origin point of faces:

- in face 4 is placed at (l; 0; 0);
- in face 6 is placed at (0; 0; 0).

For each face, original point is equal to the programmed point at position (0;0;0). Following the description of how changes the coordinate of a generic point referred to a face. For example face 1, with piece dimensions (l=1000; h=800; s=20):

- X point coordinate will be positive by moving to the right respect the origin, along the direction describe by red arrow of X axis, unlike negative values placed the point in the opposite side, to the left of the Y axis of the face;
- Y point coordinate will be positive by moving along the direction describe by green arrow of Y axis, unlike negative values placed the point in the opposite side, under the X axis of the face;
- Z point coordinate will be positive by moving to the up respect the origin, along the direction describe by blue arrow of Z axis, unlike negative values placed the point in the opposite side, under XY plane created by the face;

This means a point placed in the center of XY plane inside the piece with 10 mm of depth respect the XY face plane has coordinates X=500; Y=400; Z=-10. Same point, but outside the piece will has same XY coordinates but reverse Z, that is Z=10.

For the most uses, Z values are negative for point inside the piece, unlike positive values for points outside the piece; this true for all faces, real or not.

But it's possible to use the convention exactly reverse for the one describe above.

This document assume that the convention is first one described (most use).

CAD can be configured to change the local systems of real faces, by moving origin in XY face plane on a different corner and by change orientation of axis X and Y. In this case CAD works with a custom geometry of the piece.

Program format managed by CAD Tpa (TCN Program)

A program able to be opened with CAD Tpa is a text file with a specific format. Default extension is (.tcn).

Following there's the description of the format for information and base working managed, in order to easily interface for an external program has to produce CAD programs.

Take a look a simplified structure of program:

```
TPA\ALBATROS\EDICAD\01.00
$=test of interface cad tpa
::UNm DL=1000 DH=800 DS=40
OFFS{
#0=100
#1=50
#2=-5
}OFFS
VARV{
#0=1
#1=2
}VARV
VAR{
#0=12|diameter|w|f|fi
#1=5||w|i|tool
#2=abc||w|$|text
}VAR
SIDE#1{
$=base
W#81{ ::WTP
#1002=10 #1=101 #2=102 #3=-15 #8015=0 #2005=1.5 #2002=3300 #9012=-5 #9013=-10
#1001=1 }W
}SIDE
SIDE#2{
$=bottom
}SIDE
SIDE#3{
$=front
}SIDE
SIDE#4{
$=queue
}SIDE
SIDE#5{
$=back
}SIDE
SIDE#6{
$=head
}SIDE
```

Structure here show it's a block piece (base geometry), with minimal data set (dimensions, comment and variables) and a hole placed in face 1.

Header Lines

Take a look of blocks defined the structure.

```
TPA\ALBATROS\EDICAD\01.00
$= test of interface cad tpa
::UNm DL=1000 DH=800 DS=40
```

First row is required, for the preliminary start on program loading.

- **TPA\ALBATROS\EDICAD** Is a constant string
- **xx:xx:xxx** represent the version of the cad
 - 00: EDICAD
 - 01: TPAEDI32
 - 02: TPACAD
- **"r"** password level for reading of a TCN file
- **"w"** password level for writing of a TCN file.
- **"h" ?????.**
 - 0= low
 - 1= test
 - 2= high password
- **"s"** number of savings. (from 4.0, is used to check if program has been saved since the last use)
- **`tcn version=2.4.0:**
- **`code=ansi:**

Second row, start with "\$=" set the program comment and is optional. If exists, has to start with prefix "\$=", followed by the description.

::SIDE=1;... defines the faces that include operations

Third row is necessary and sets measure unit and dimensions:

- **“:.”** is the header
- **“UNm”** measured unit in [mm] (default); **“UNi”** measured unit in [inch]
- **“DL=1000 DH=800 DS=40”** dimensions: DL= length, DH= height, DS= thickness.

Fields are separated by space char.

Execution section

```

EXE{
#0=1
#1=1
#2=0
#3=0
#4=0
}EXE

```

#0	Execution mode, 0= normal, 1= mirror in X, affects the setting mirror in CNC Board
#1	Work area, 0=N; 1=M; 2=S; 5=R; 6=N1;7=M1, area definition if programm is called in CNC Board
#2	Locator offset X; not managed
#3	Locator offset Y; not managed
#4	Locator offset Z; not managed

Variables section

```

OFFS{
#0=100
#1=50
#2=-5
}OFFS
VARV{
#0=1
#1=2
}VARV
VAR{
#0=12|diameter|w|f|fi
#1=5||w|i|tool
#2=abc||w|$|text
}VAR

```

Sections are optional and could set global variables of the program. Like describe following, sections could be three, each one referred to a specific type of variables:

- OFFS{ ... }OFFS: variables “o” section. “o” variables could be maximum 8 (in this example only 3).
- VARV{ ... }VARV: variables “v” section. “v” variables could be maximum 8 (in this example only 2).
- VAR{ ... }VAR: variables “r” section. “r” variables could be maximum 300 (in this example only 3).

Variables “o” and “v”

```

OFFS{
#0=100
#1=50
#2=-5
}OFFS
VARV{
#0=1
#1=2
}VARV

```

The sections of these two groups of variables have the same structure. Take a look of “o” variables.

The first and the last row are required, to open and close the section: “OFFS{“, “}OFFS”.

Each internal row describes a single variable:

#0= for variable o0 (accepted rows from “#0=” to “#7=”)
 100 numeric values set to the variable

Variables “r”

```

VAR{
#0=12|diameter|w|f|fi
#1=5||w|i|tool
#2=abc||w|$|text
}VAR

```

The first and the last row are required, to open and close the section: "VAR{", "{}VAR".

Each internal row describes a single variable:

#0=	for variable r0 (accepted rows from: "#0=" to "#299=")
12	numeric values set to the variable (maximum length: 100 characters)
diameter	comment text of variable (maximum length: 100 characters, space chars replaced with ~ char). Field can be empty.
w	visibility state (r: private, w: public) . If not exists it considered: <i>private</i> .
f	variable type (f: double, i: integer, \$: string). If not exists it considered: double.
fi	symbolic name of variable (maximum length: 16 characters, lowercase, alphanumeric), Field can be empty.
	field separator

Face Program Section

```

SIDE#1{
$=base
W#81{ ::Wtp
#1002=10 #1=101 #2=102 #3=-15 #8015=0 #2005=1.5 #2002=3300 #9012=-5 #9013=-10
#1001=1 }W
}SIDE

```

First row is required, necessary to open section of face program: "SIDE#1{" open section face 1, ..., "SIDE#6{" open section face 6.

Second row, starting with "\$=" set the face name and it is optional. If exist, have to be start with prefix "\$=", followed by the face name.

Following there're block starting with "W#nn{ ::" and closing with "}W", to define the existing workings on the face.

Last row ("}SIDE") is required, to close the face section.

If the workpieceface, face #0 is selected in addition the definition of the application face has to be assigned: WF=....

e.g

```

SIDE#0{
$=F #0
W#89{ ::WTs WF=3 #8015=0 #1=100 #2=100 #3=-s+50 #201=1 #203=1 #205=1612
#1001=100 #9521=0 #8101=0 #8096=0 #40=0 #46=1 #8135=0 #8136=0 #8180=0
#8181=0 #8185=0 #8186=0 #9511=0 #9512=0 #9513=0 }W
This means the setup is executed on face #3.

```

It's not necessary to set sections for those faces not have any programmed workings inside them. Faces sections have to be at the end of the text file.

Working section assigned to face program

```
W#81{ ::WTP
#1002=10 #1=101 #2=102 #3=-15 #8015=0 #2005=1.5 #2002=3300 #9012=-5 #9013=-10
#1001=1 }W
```

A working can be placed on one or more text lines. Following some examples of the same working:

```
W#81{ ::WTP #1002=10 #1=101 #2=102 #3=-15 #8015=0 #2005=1.5 #2002=3300 #9012=-5
#9013=-10 #1001=1 }W
```

```
W#81{ ::WTP
#1002=10 #1=101 #2=102 #3=-15 #8015=0 #2005=1.5 #2002=3300 #9012=-5 #9013=-10
#1001=1
}W
```

```
W#81{ ::WTP #1002=10 #1=101 #2=102 #3=-15
#8015=0 #2005=1.5 #2002=3300 #9012=-5 #9013=-10 #1001=1
}W
```

```
W#81{ ::WTP
#1002=10 #1=101 #2=102 #3=-15
#8015=0 #2005=1.5 #2002=3300 #9012=-5 #9013=-10 #1001=1
}W
```

where the working elements is placed on same line or split in more lines, keeping the congruence of following syntax rules:

- fields in same line is separated by space character;
- section header have the same fixed structure “W#nn{ ::WTc WS=” (example: “W#81{ ::WTP WS=10”), with:
 - W#nn= working operation code (numeric),
 - WTc= char describe working type
 - ‘p’=punctual,
 - ‘s’=setup,
 - ‘l’=line,
 - ‘a’=arc,
 - ‘2’=complex
 - WS= continues number of programming (it is not necessary to assigne;it is assignet automatically, when the program is saved)
 - WB= Construction setting (if =1 then the line is not executed, not present if =0)
 - WS= ATTENTION not in use (exclution of working, comment)
- remain fields have fixed structure “#nn=st” (example: “#1002=10”), with:
 - nn= numeric identify of parameter,
 - st= value set to parameter;
 (a single field can’t be split in more than one line)
- section closing has fixed structure “}W”.

Working: Hole

The header is: "W#81{ ::WTP".

Geometric parameters:

#1=	X dimension of application
#2=	Y dimension of application
#3=	Depth dimension in Z
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions

If a coordinate not set will be taken the value of that coordinate from the previous work in the working list.

Technological parameters in case of programming hole by diameter:

#1002=	Tool Diameter
#201=	Machine
#203=	Group
#1001=	Tool Type

Technological parameters in case of programming hole by tool:

#201=	Machine
#203=	Group
#205=	Tool
#1001=	Tool Type

Generic technological parameters:

#2005=	Entry tool speed (m/min.)
#2002=	Rotation speed of the spindle (rev/min.)
#9012=	Position of slowing down speed in tool entry
#9013=	Position of slowing down speed in tool exit

If working set both the tool (#205=) and the diameter (#1002=): the tool has the priority and the diameter set will be ignored.

Working: Setup mill

```
W#89{ ::WTs WS=5 #8015=0 #1=100 #2=100 #3=-10 #201=1 #203=1 #205=5408
#1001=100 #9521=0 #8101=0 #8096=0 #40=0 #46=1 #8135=0 #8136=0 #8180=0
#8181=0 #8185=0 #8186=0 #9511=0 #9512=0 }W
```

The header is: "W#89{ ::WTs".

Geometric parameters:

#1=	X dimension of application
#2=	Y dimension of application
#3=	Depth dimension in Z
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions

If a coordinate not set will be taken the value of that coordinate from the previous work in the working list.

Technological parameters for tool selection:

#201=	Machine (has to be =1)
#203=	Group (has to be =1)
#205=	Tool
#1001=	Tool type (has to be set to =100)

Generic technological parameters:

#2005=	Entry tool speed (m/min.)
#2002=	Rotation speed of the spindle (rev/min.)
#40=	Tool compensation application: 1 for left compensation, 2 for right compensation, 0 for disable the compensation(default)
#36=	Compensation radius(if not set: will be used the radius of the tool)
#46=	Reduces the profile
#8135-#9191=	Defintion of entry and exit arc see TPACAD manual
#9521=	Rotationdirection: 0=default, 1=left, 2 right
#8096=	1= Multisetup

Properties:

#9511=	If =1 defined as nesting geometrie
#9512=	If =1 defined as leftover area for nesting calculation

Working: Line

| W#2201{ ::WTI #1=100 #2=400 #3=-15 }W

The header is: "W#2201{ ::WTI".

Geometric parameters:

#1=	X dimension of application (ending point of the line)
#2=	Y dimension of application
#3=	Depth dimension in Z
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions

If a coordinate not set will be taken the value of that coordinate from the previous work in the working list.

Generic technological parameters:

#2008=	Interpolation Speed (m/min.)
--------	------------------------------

Working: Arc in face plane

| W#2101{ ::WTa #1=100 #2=400 #31=80 #3=-15 }W

The header is: "W#2101{ ::WTa".

Geometric parameters:

#1=	X dimension of application (ending point of the arc)
#2=	Y dimension of application
#3=	Depth dimension in Z
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions
#31=	Center X position, <u>relative</u> respect the X position of the arc initial point
#32=	Center Y position, <u>relative</u> respect the Y position of the arc initial point
#34=	Rotation Arc Sense: 0 if clockwise(default), 1 if counter-clockwise

If a coordinate (XYZ) not set will be taken the value of that coordinate from the previous work in the working list.

Generic technological parameters:

#2008=	Interpolation speed (m/min.)
--------	------------------------------

Note

Arc parameters defined and described above are "iperstatic", that is too many to define an arc in a free way, so it's possible to define only one coordinate X or Y about the arc center. In this way the other coordinate is calculated automatically without user set redundancy information. Like generic rule, it's recommended to set only Cx value if the distance from Xend and Xstart is less than the distance from Ystart and Yend, and to set only Cy otherwise. This convention is required in case of Xend – Xstart = 0 (Cx has to be set) or Yend – Ystart = 0 (Cy has to be set). Only in case of complete circle (starting point equal to ending one) must be set both coordinates of center point.

Working: Saw X

W#1050{ ::WT2 WS=1 #8098=..\custom\mcr\lame.tmc #6=1 #8503=0 #8509=0
 #8510=0 #8511=10 #8512=-10 #8514=1 #8515=1 #8516=1161 #8517=100 #8525=0
 #8526=0 #8527=0 }W

The header is: "W#1050{ ::WT2".

The working can be usually applied on face 1

Geometric parameters:

#8098=	Macro that is used (has to be #8098=..\custom\mcr\lame.tmc)
#8510=	Position X of starting application point
#8517=	Position X of ending application point
#8511=	Position Y of application
#8512=	Position Z of depth
#8509=	Sawtype (has to be set to =1)
#6=	Selection of the face (has to be set to =1)
#8503=	Width of saw-cut (if only one cut = 0, if wider than the thickness of the blade enter required width of the cut)
#8504=	Has to be set =subang

Technological parameters for tool selection:

#8514=	Machine (has to be =1)
#8515=	Group (has to be =1)
#8516=	Tool ID

Generic technological parameters:

#8522=	Rotation speed of the spindle (rev/min.) (not necessary because defined by tool-parameter)
#8524=	Entry tool speed (m/min.) (not necessary because defined by tool-parameter)
#8523=	Executing saw speed (m/min.) (not necessary because defined by tool-parameter)

Specific technological parameters of the saw:

#8527=	If =1 is been calculated the bow-string for the starting point
#8525=	Tool compensation application: 1 for left compensation, 2 for right compensation, 0 for disable the compensation(default)
#8526=	If =1 saw will be executed in two step

#8513=	Position Z of depth of the second step of the working (not necessary if 8526 not =1)
#8529=	Executing saw speed of the second step (m/min.) (not necessary because defined by tool-parameter)

Working: Saw Y

W#1051{ ::WT2 WS=2 #8098=..\custom\mcr\lame.tmc #6=1 #8503=10 #8509=1
 #8510=100 #8511=0 #8512=-10 #8514=1 #8515=1 #8516=1162 #8518=100 #8525=0
 #8526=0 #8527=0 }W

The header is: "W#1051{ ::WT2".

The working can be usually applied on face 1

Geometric parameters:

#8098=	Macro that is used (has to be #8098=..\custom\mcr\lame.tmc)
#8510=	Position X of starting application point
#8518=	Position Y of ending application point
#8511=	Position Y of application
#8512=	Position Z of depth
#8509=	Sawtype (has to be set to =1)
#6=	Selection of the face (has to be set to =1)
#8503=	Width of saw-cut (if only one cut = 0, if wider than the thickness of the blade enter required width of the cut)
#8504=	Has to be set =subang

Technological parameters for tool selection:

#8514=	Machine (has to be =1)
#8515=	Group (has to be =1)
#8516=	Tool ID

Generic technological parameters:

#8522=	Rotation speed of the spindle (rev/min.) (not necessary because defined by tool-parameter)
#8524=	Entry tool speed (m/min.) (not necessary because defined by tool-parameter)
#8523=	Executing saw speed (m/min.) (not necessary because defined by tool-parameter)

Specific technological parameters of the saw:

#8527=	If =1 is been calculated the bow-string for the starting point
#8525=	Tool compensation application: 1 for left compensation, 2 for right compensation, 0 for disable the compensation(default)

#8526=	If =1 saw will be executed in two step
#8513=	Position Z of depth of the second step of the working (not necessary if 8526 not =1)
#8529=	Executing saw speed of the second step (m/min.) (not necessary because defined by tool-parameter)

Working: Saw XY

```
W#1052{ ::WT2 WS=3 #8098=..\custom\mcr\lame.tmc #6=1 #8503=0
#8504=subang #8507=10 #8508=1 #8509=2 #8510=0 #8511=0 #8512=-10 #8514=1
#8515=1 #8516=1236 #8519=45 #8520=200 #8521=90 #8525=0 #8526=0 #8527=0
#8531=0 #8532=0 }W
```

The header is: "W#1051{ ::WT2".

The working can be usually applied on face 1

Geometric parameters:

#8098=	Macro that is used (has to be #8098=..\custom\mcr\lame.tmc)
#8510=	Position X of starting application point
#8511=	Position Y of application
#8512=	Position Z of depth, value is in direction of BETA angle and not perpendicular to face#1. Can be changed by value 8508
#8509=	Sawtype (has to be set to =2)
#6=	Selection of the face (has to be set to =1)
#8503=	Width of saw-cut (if only one cut = 0, if wider than the thickness of the blade enter required width of the cut)
#8507=	Application layer for Z position (not necessary if not used)
#8533	Enable X/Y final position (modul and alfa angle are auto. dissabled)
#8517	Final X position
#8518	Final Y position
#8508=	Selection Z abs (if 0= not active, if 1= acitive)
#8519=	Angle Alfa, direction of the saw cut, direction X=0°, direction Y=90°
#8521=	Angle Beta, tilting of the sawblade
#8520=	Length of the cut
#8504=	Has to be set =subang

Technological parameters for tool selection:

#8514=	Machine (has to be =1)
#8515=	Group (has to be =1)

#8516=	Tool ID
--------	---------

Generic technological parameters:

#8522=	Rotation speed of the spindle (rev/min.) (not necessary because defined by tool-parameter)
#8524=	Entry tool speed (m/min.) (not necessary because defined by tool-parameter)
#8523=	Executing saw speed (m/min.) (not necessary because defined by tool-parameter)

Specific technological parameters of the saw:

#8527=	If =1 is been calculated the bow-string for the starting point
#8525=	Tool compensation application: 1 for left compensation, 2 for right compensation, 0 for disable the compensation(default)
#8526=	If =1 saw will be executed in two step, depth defined in 8513 will be executed reverse feeding direction
#8513=	Position Z of depth of the second step of the working (not necessary if 8526 not =1)
#8529=	Executing saw speed of the second step (m/min.) (not necessary because defined by tool-parameter)

Properties:

#8531=	If =1 defined as nesting geometrie
#8532=	If =1 defined as leftover area for nesting calculation

Working: Subprogram application

```
W#2010{ ::WT2 #8098=a.tcn #6=1 #8015=0 #8020=100 #8021=100 #8022=-12 #8044=30
#8103=1 #8104=0 }W
```

The header is: "W#2010{ ::WT2".

Geometric parameters:

#8098=	Subprogram name (name.extension). Can be set a path too, referred to the directory of subprograms. All space characters have to be replaced with ~ character
#8020=	Position X of application of subprogram
#8021=	Position Y of application
#8022=	Position Z of application
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions

If a coordinate (XYZ) not set will be taken the value of that coordinate from the previous work in the working list.

Subprograms variables:

#6=	Number of the face of subprogram has to be call.
#8500=	Set r0 variable of subprogram (has to be defined: public)
..	...
#8800=	Set r299 variable of subprogram (has to be defined: public)

Parameters of geometric transformations:

#8044=	Rotation angle.
#8103=	If =1 mirror respect to an axis parallel to Y face axis.
#8104=	If =1 mirror respect to an axis parallel to X face axis.

Working: Geometric Set-up G203 for 3-4 axis machines

```
W#203{ ::WTs WS=7 #8015=0 #1=100 #2=10 #3=-s-5 #201=1 #203=1 #205=1614 #1001=100
#5=-23 #4=90 #9521=0 #47=1 #9514=2 #9511=0 #9512=0 #9513=0 #9515=-s #9516=1 #9517=2
}W
```

The header is: "W#203{ ::WTs".

Geometric parameters:

#1=	X dimension of application
#2=	Y dimension of application
#3=	Depth dimension in Z
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions

If a coordinate not set will be taken the value of that coordinate from the previous work in the working list.

Technological parameters for tool selection:

#201=	Machine (has to be =1)
#203=	Group (has to be =1)
#205=	Tool
#1001=	Tool type (has to be set to =100)
#5=	Beta angle (0°= perpendicular to the face, -90° parallel to the face)
#4=	Alfa angle (0° towards X +, 90° toward Y +)
#9521=	Rotationdirection: 0=default, 1=left, 2 right
#47=1	This value is not displayed in TPACAD but has to be set to 1
#9514=2	This value is not displayed in TPACAD but has to be set to 1

Generic technological parameters:

#2005=	Entry tool speed (m/min.)
#2002=	Rotation speed of the spindle (rev/min.)

Properties:

#9511=	If =1 defined as nesting geometrie
#9512=	If =1 defined as leftover area for nesting calculation
#9515=-s	Z value of application layer

#9516=1	If =1 the Z value defined is absolute and not in direction of beta angle
#9517=2	If=0 not active, if=1 correction above, if=2 correction below

Working: Geometric Set-up G205 for 3-4 axis machines (setup with chip-deflector)

W#205{ ::WTs WS=1 #1=-100 #2=0 #3=-s #201=1 #203=1 #205=1600 #1001=100 #5=geo[beta]
 #4=0 #9521=0 #40=2 #9513=1 }W

The header is: "W#205{ ::WTs".

Geometric parameters:

#1=	X dimension of application
#2=	Y dimension of application
#3=	Depth dimension in Z
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions

If a coordinate not set will be taken the value of that coordinate from the previous work in the working list.

Technological parameters for tool selection:

#201=	Machine (has to be =1)
#203=	Group (has to be =1)
#205=	Tool
#1001=	Tool type (has to be set to =100)
#5=	Beta angle has to be set to geo[beta] . Can not be changed
#4=	Alfa angle (0° towards X +, 90° toward Y +)
#9521=	Rotationdirection: 0=default, 1=left, 2 right
#40=	Tool compensation application: 1 for left compensation, 2 for right compensation, 0 for disable the compensation(default)
#36=	Compensation radius(if not set: will be used the radius of the tool)

Generic technological parameters:

#2005=	Entry tool speed (m/min.)
#2002=	Rotation speed of the spindle (rev/min.)

Properties:

#9513=	If=1 use of chip deflector
--------	----------------------------

Working: Geometric Set-up G206 for 3-4 axis machines (oriented setup in lateral faces)

W#206{ ::WTs WS=2 #8015=0 #1=20 #2=15 #3=-30 #201=1 #203=1 #205=9500 #1001=100
 #5=-45 #4=geo[alfa] #9521=0#40=0 }W

The header is: "W#206{ ::WTs".

Geometric parameters:

#1=	X dimension of application
#2=	Y dimension of application
#3=	Depth dimension in Z
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions

If a coordinate not set will be taken the value of that coordinate from the previous work in the working list.

Technological parameters for tool selection:

#201=	Machine (has to be =1)
#203=	Group (has to be =1)
#205=	Tool
#1001=	Tool type (has to be set to =100)
#5=	Beta angle (0°= perpendicular to the face, -90° parallel to the face)
#4=	Alfa angle (0° towards X +, 90° toward Y +)
#9521=	Rotationdirection: 0=default, 1=left, 2 right
#9514=2	This value is not displayed in TPACAD but has to be set to 1
#40=	Tool compensation application: 1 for left compensation, 2 for right compensation, 0 for disable the compensation(default)
#36=	Compensation radius(if not set: will be used the radius of the tool)

Generic technological parameters:

#2005=	Entry tool speed (m/min.)
#2002=	Rotation speed of the spindle (rev/min.)

Working: Geometric Set-up G203 for 5 axis machines

```
W#203{ ::WTs WS=7 #8015=0 #1=100 #2=10 #3=-s-5 #201=1 #203=1 #205=1614 #1001=100
#5=-23 #4=90 #9521=0 #47=1 #9514=2 #9511=0 #9512=0 #9513=0 #9520=0 #9515=-s
#9516=1 #9517=2 }W
```

The header is: "W#203{ ::WTs".

Geometric parameters:

#1=	X dimension of application
#2=	Y dimension of application
#3=	Depth dimension in Z
#8015=	Application dimensions are expressed in Absolute = 0 (default)/ Relative = 1. Applied to all XYZ dimensions

If a coordinate not set will be taken the value of that coordinate from the previous work in the working list.

Technological parameters for tool selection:

#201=	Machine (has to be =1)
#203=	Group (has to be =1)
#205=	Tool
#1001=	Tool type (has to be set to =100)
#5=	Beta angle (0°= perpendicular to the face, -90° parallel to the face)
#4=	Alfa angle (0° towards X +, 90° toward Y +)
#9521=	Rotationdirection: 0=default, 1=left, 2 right
#47=1	This value is not displayed in TPACAD but has to be set to 1
#9514=2	This value is not displayed in TPACAD but has to be set to 1

Generic technological parameters:

#2005=	Entry tool speed (m/min.)
#2002=	Rotation speed of the spindle (rev/min.)

Properties:

#9511=	If =1 defined as nesting geometrie
#9512=	If =1 defined as leftover area for nesting calculation
#9513=0	If=1 use of chip deflector

#9520=	Enable Z Layer application (9515), 0= Zlayer is not applied, 1= Zlayer active
#9515=	Z value of application layer
#9516=	If =1 the Z value defined is absolute and not in direction of beta angle
#9517=	If=0 not active, if=1 correction above, if=2 correction below

Special function: Rapid setup G102

This function is used to interrupt the program execution. Machine is moving to the defined position and green button starts to blink. If the button is pushed the machine continues with the execution of the program

W#102{ ::WTg WF=1 #1=100 #2=100 #3=-10 #201=1 #205=100 #9505=99 }W

The header is: "W#102{ ::WTg".

(Attention: W102 is defined as Setup in machine H10 in connection with label-printing and can not be used as rapid in this case; the setting in this case is: "WTs" instead of "WTg")

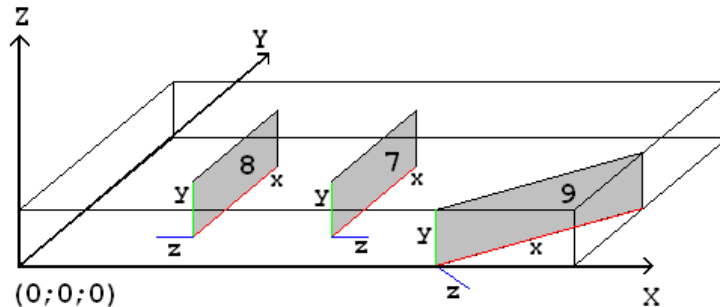
Geometric parameters:

#1	X dimension of application. In relation to the zero-point of piece and the defined tool (#205)
#2	Y dimension of application. In relation to the zero-point of piece and the defined tool (#205)
#3	Depth dimension in Z. Not used, machine automatically moves to Z=0 (all the way up)
#201=	Machine (default =1, not necessary)
#203=	Group (default = 1, not necessary)
#204	EM (not necessary)
#205	Tool-number, the tool has to be part of the outfit that is active. Default is tool #1 (first drill)
#9505	Function definition 98: machine moves in defined position and releases the vacuum of the piece. Restart with green button 99: machine moves in to the defined position and vacuum is not released. Restart with green button

Programming fictitious faces

Inside piece can be added new faces with whatever orientation: fictitious faces. Fictitious faces have an identification number from 7 to 99. Enumeration of faces can be not consecutive.

Let's see an example, underline that the example simplify at maximum the way to set an oriented plane in CAD Tpa:



The figure above is an example of a program with three fictitious faces: with face numbers from 7 to 9.

Faces plane is painted in grey in order to highlight them and are show: face numbers and axis labels of the face (xyz), on the side of positive programming.

The axis triad with the origin in (0;0;0) show in the left-bottom corner of the figure is the absolute reference system to which are referred the corners of each fictitious face.

Following example highlight parts of the file that set the three new fictitious faces:

```
TPA\ALBATROS\EDICAD\01.00
$=test interface cad tpa
::UNm DL=1000 DH=800 DS=40
....
GEO{ ::NF=3
  GSIDE#7{
    #1=400|100|0
    #2=400|300|0
    #3=400|100|18
    #Z=18
  }GSIDE
  GSIDE#8{ ::AZ=1
    #1=200|100|0
    #2=200|300|0
    #3=200|100|18
    #Z=18
  }GSIDE
  GSIDE#9{
    #1=600|0|0
    #2=800|200|0
    #3=600|0|18
    #Z=18
  }GSIDE
}GEO
  SIDE#1{
```

Section is placed before sections of programming face.

Section is described between open and close lines (write in bold):

- "GEO{ ::NF=3" open the section and set the total number of fictitious faces (three)
- "}GEO" close the section

Follow each section for all fictitious face defined (in the example: three):

- "GSIDE#7{" open section of face 7 and "}GSIDE" close it
- "GSIDE#8{" open section of face 8 e "}GSIDE" close it
- "GSIDE#9{" open section of face 9 e "}GSIDE" close it

Fictitious face section

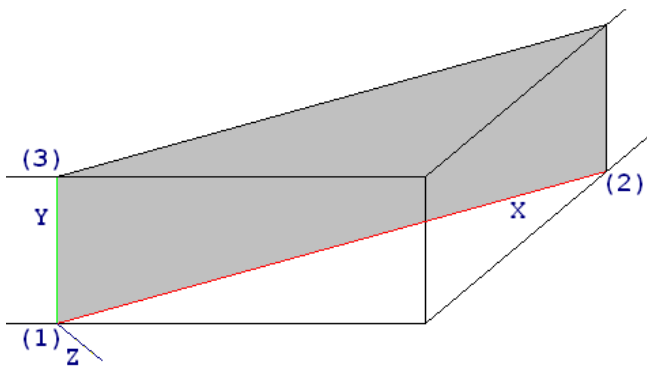
```

| GSIDE#8{ ::AZ=1
| #1=200|100|0
| #2=200|300|0
| #3=200|100|18
| #Z=18
| }GSIDE

```

In the example of face 8, section header ends with “::AZ=1” (lack for other two face), to define a left-hand triad for the face.

#1=200 100 0	First face corner: is the origin of face axis, that is the programmed point (0;0;0): “#1=” identify the corner (1) “200 “ X point coordinate “100” Y point coordinate “0” Z point coordinate “ ” field separator
#2=200 300 0	Second face corner: is the boundary point in positive direction along X axis
#3=200 100 18	Third face corner: is the boundary point in positive direction along Y axis
#Z=18	Face Thickness. This row is not obligatory, if not present, face thickness will be equal to piece thickness.



See in details face 9 of the example: figure on the left show its three corners:

- Distance from corner (1) and (2) is the length of the face
- Distance from corner (1) and (3) is the height of the face

In this example all corners are programmed in order to obtain a 90° angle. If not like this: Y axis of the face will be automatically converted in order to put X and Y axis perpendicular.

OPTION: Lablemanagment

SIDE#1{
W#102{ ::WTs WS=1 #1=111 #2=222 #3=0 #201=1 #205=1 #9505=100 }W

The header is: "W#102{ ::WTs".

The lableposition has to be defined using the function W#102.

#1	X point coordinate of lable
#2	Y point coordinate of lable
#3	Z point coordinate of lable (=zero)
#201	Machine (has to be =1) default=1
#203	Group (has to be =1) default=1
#204	SubGroup (has to be =1) default =1
#205	Tool (has to be =1)
#9505	Function (has to be =100)
#9508	M1 field (default =0) not used
#9509	M2 field (default =0) not used
#95010	M3 field (default =0) not used

DXF Import

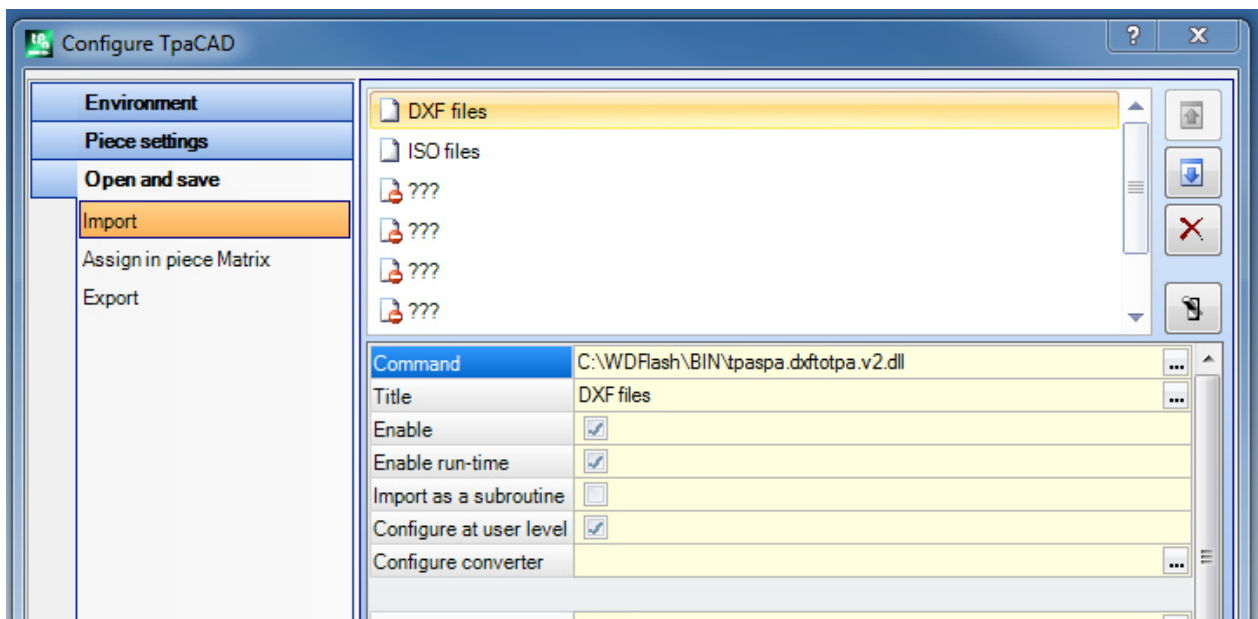
For the DXF Import and the necessary settings see the tpacad manual
Chapter: 15.1 From Dxf to TpaCAD format

ATTENTION:

Depending on the settings in TPACAD two versions of the importer are available

Default for 3.2 and 4.0: TpaSpa.DxfToTpa.v2.dll

Default for 4.1: TpaSpa.DxfCad.v2.dll



ISO code direct import with conversion in *.tcn

For the ISO code definition see the tpaCAD manual
Chapter: 15.3 From ISO to TpaCAD format

This function is only available if in the TPACAD configuration following DLL is activated
TpaSpa.IsoToTpa.v2.dll

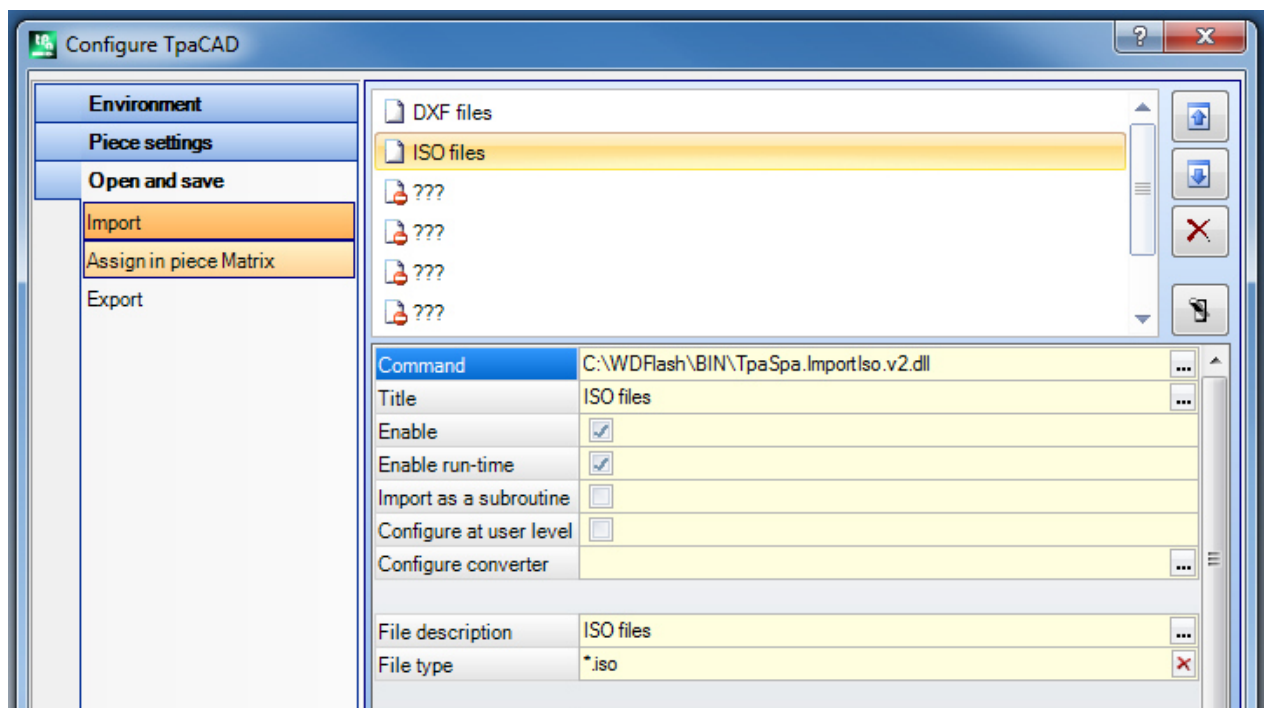
Additional information

- For this kind of import there are following limits:
 - o Only in face#1
 - o Only lines and holes
 - o No 5axis movement
- The functionality is that the iso file is converted internally to a *.tcn file so this application does not solve the problem of very big programmes (can not be compared with the 5 axis iso-code function)

ATTENTION:

The default setting in 4.0,4.1 is the dll
TpaSpa.ImportIso.v2.dll

With this dll the import always generates a SUBISO call (see next chapter)

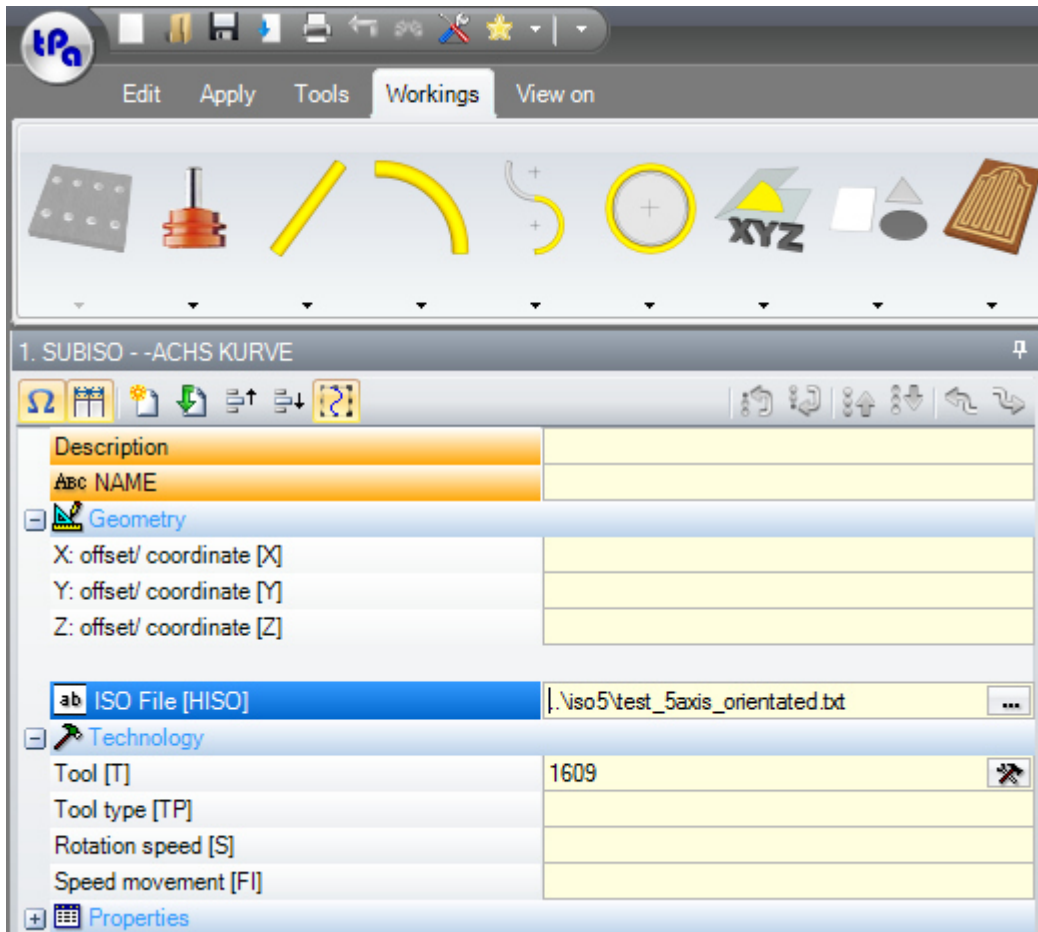


isocode call in TPACAD (for 5 axis and also 3 axis applications)

General 5axis isocode:

In this case a very simple iso file is generated. This file is saved in a defined path. In TpaCAD there is a function called SUBISO. With this function the name and path of the iso file is defined and which tool has to be used

This command can also be used in machines that have only 3 axis. In this case the setting 5AX has to be set to zero and the ISO code must not contain C and A axis values.



The SUBISO call can only be applied in face#1 (or in the face#0 calling application face #1). A shift of the coordinate system using fictive faces is not possible, the values always revert to the coordinate system of the piece. (attention Z =0 is the bottom of the piece!)

Structur ISO file for 5Axis curvs (SUBISO call)

G0: this is the starting point for the milling. (rapid movement is done until this point is reached)

G1: Micro-lineare interpolations movements

G2,G3: a segment of circular interpolation

X, Y, Z value: Always absolute measurements in mm referred to the center of the tool tip, with the zero point the SUBISO X Y Z call coordinate. (the zero point is the left bottom corner of the workpiece, Z=0 level of the cups!)

C and A value: absolute angle measurement of the axis in degree, Starting from the required direction of the tool the absolute axis values for the C and A axis have to be calculated using the 50° prisma-head. It is not necessary to respect the mechanic offsets, this is done automatically by the control

F: feeding speed in mm/min (this value is necessary otherwise a very slow default speed is applied)

M2: has to be at the end of an iso file

If parameters are not defined they are zero or assumed from the previous line
The tool is defined in the TpaCAD program.

Following regulations:

- Capital letters have to be used.
- Between all informations a space has to be used (G0_X100_Y200...)

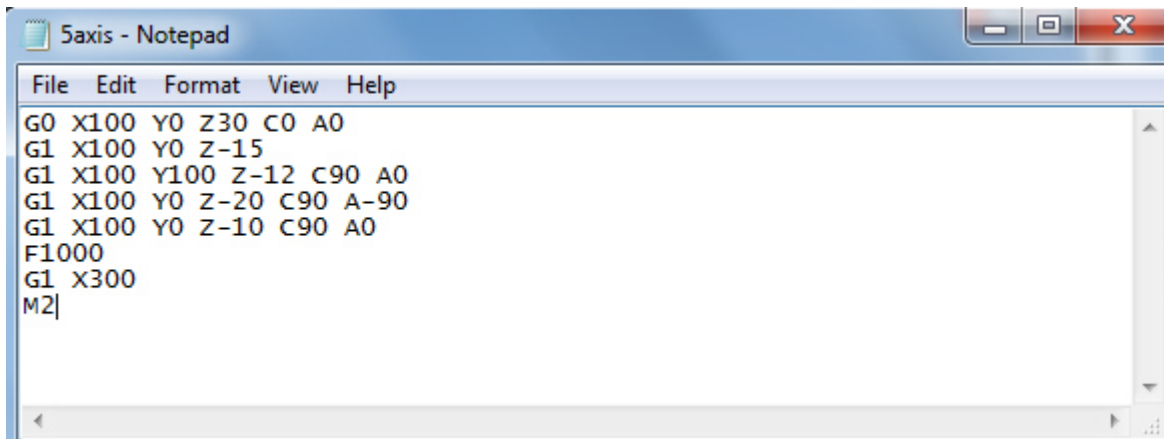
RTCP Function: the clearance movement between two setups is calculated by the PLC if the program is coming from TPACAD. Also in the case that the A and C Axis have to be rotated the Z axis position is high enough to prevent collisions.

If the program is coming from an iso file it is recommended to set in CNC Board under the tab “main switch” the selection “move A and C axis with Z axis up”. In this case the Z axis is always moving all the way up before rotating A and C axis. In this way collisions are prevented.

In case of interpolations with fast movements of the A and the C axes, it is better to cut the profile in more segments: in this way the machine will be able to manage better the accelerations and the speeds for all the axes.

The file has to be saved in ASCII format as *.iso or in *.txt

Example:

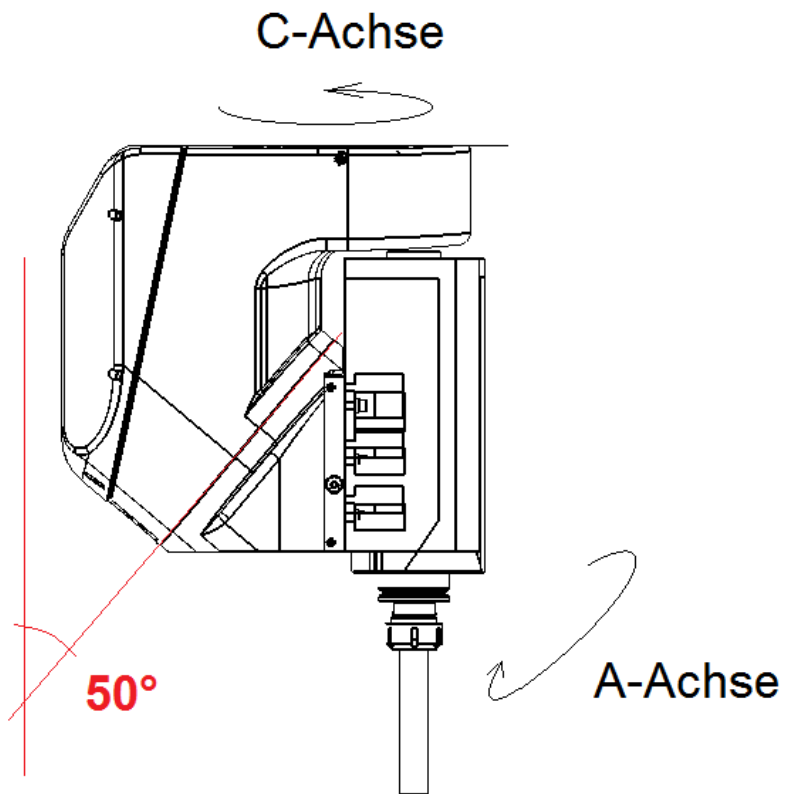


A screenshot of a Notepad window titled "Saxis - Notepad". The window has a standard menu bar with "File", "Edit", "Format", "View", and "Help". The text area contains the following G-code:

```
G0 X100 Y0 Z30 C0 A0  
G1 X100 Y0 Z-15  
G1 X100 Y100 Z-12 C90 A0  
G1 X100 Y0 Z-20 C90 A-90  
G1 X100 Y0 Z-10 C90 A0  
F1000  
G1 X300  
M2|
```

Prisma head:

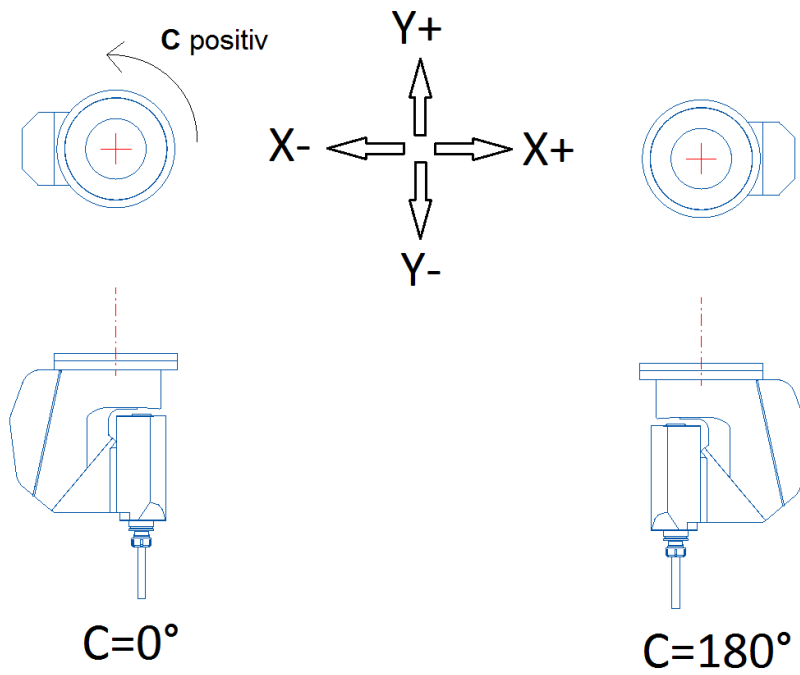
Angle between axis: 50°



C-Axis

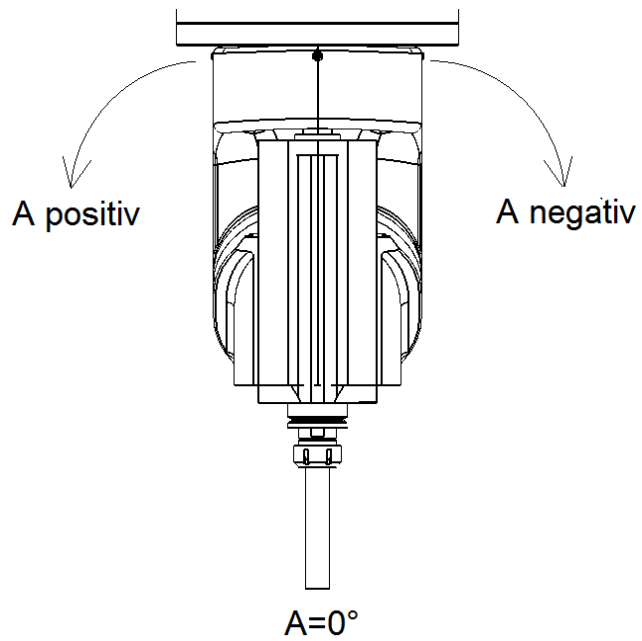
The limit for the C axis is $\pm 270,1^\circ$.

The zero position for the C axis is when the motor is on the right side of the head. Rotation in direction C positive is counterclockwise if you look on the spindle from the top.



A-axis

The limit for the A axis is $\pm 181^\circ$



Explanation saw-cut 5 axis machine

Rotation: Rotation-direction of the electro spindle

Saw rotation direction: direction of the execution of the sawcut

- 2- Clockwise:
 - View from the direction of the main spindle
 - The entry-movement , the feeding direction and the exit movement are in clockwise direction
- 1-Counterclockwise:
 - View from the direction of the main spindle
 - The entry-movement , the feeding direction and the exit movement are in counter-clockwise direction

Tool ID	5000
Comment	Nutscheibe
Saw data	
Saw Diameter (mm)	301.5
[MAXZ] Max Depth (mm)	88
[D] Saw Thickness (mm)	3.46
[LA] Fulcrum-Tip interaxis (mm)	65
Axis C Offset (°)	0
[THIC] Internal Blade Thickness	3.46
Spindle rotation	
Min. RPM	3000
Max. RPM	5500
Default RPM	5500
Rotation	1 - M04_left
Saw Rotation direction	1 - Counterclockwise
Speed value	
Default Work Speed (m/min)	8
Default Penetration Feed (m/min)	1
Other data	
Tool Acceleration Time (Seconds)	3

Change Image

Tool Profile

Change Photo...

Tool Unit

OK

Cancel

Usually the saw-blade (aggregate or on HSK cone) is defined so that the first cut [value defined in Zp] is done in scoring function.

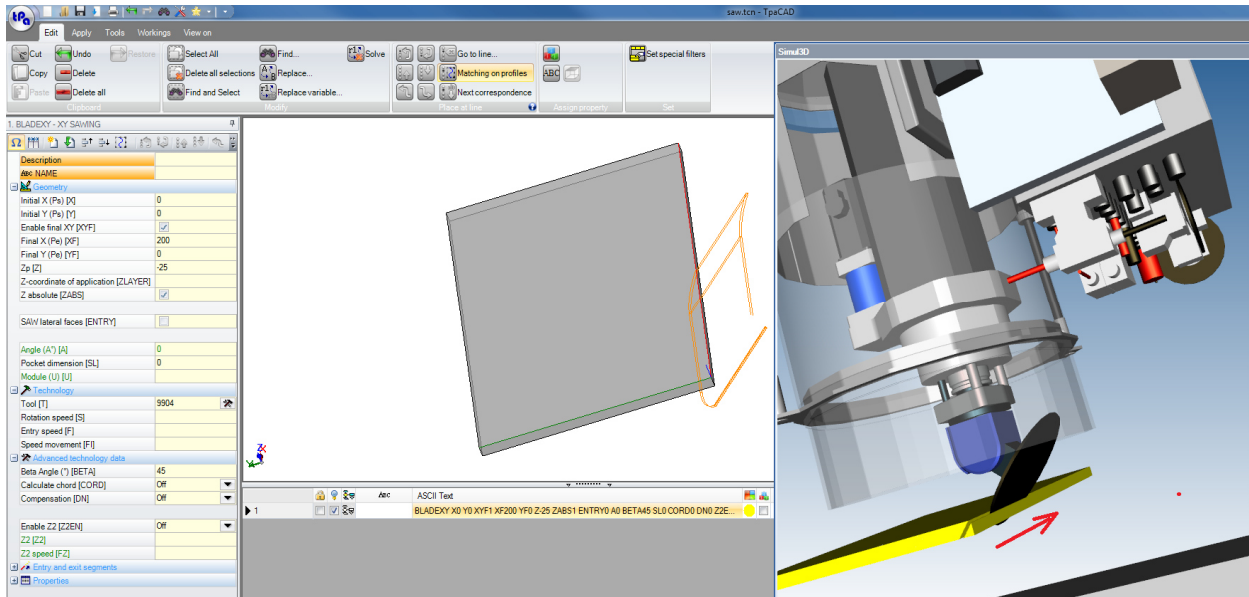
The depth defined in Z2 is executed with reversed feeding-direction. That means I can program a pre-cut and a finish cut in one line using e.g. Zp=-1 and Z2= -19. Of course there is also the option to leave the field Zp empty and only use the field Z2. In this way the feeding-direction is reversed.

If the beta angle is 90° (saw-blade perpendicular to the face #1) the feeding direction is changed regarding X and Xf but also C axis position is reversed so the relation between rotation and feeding direction remains the same.

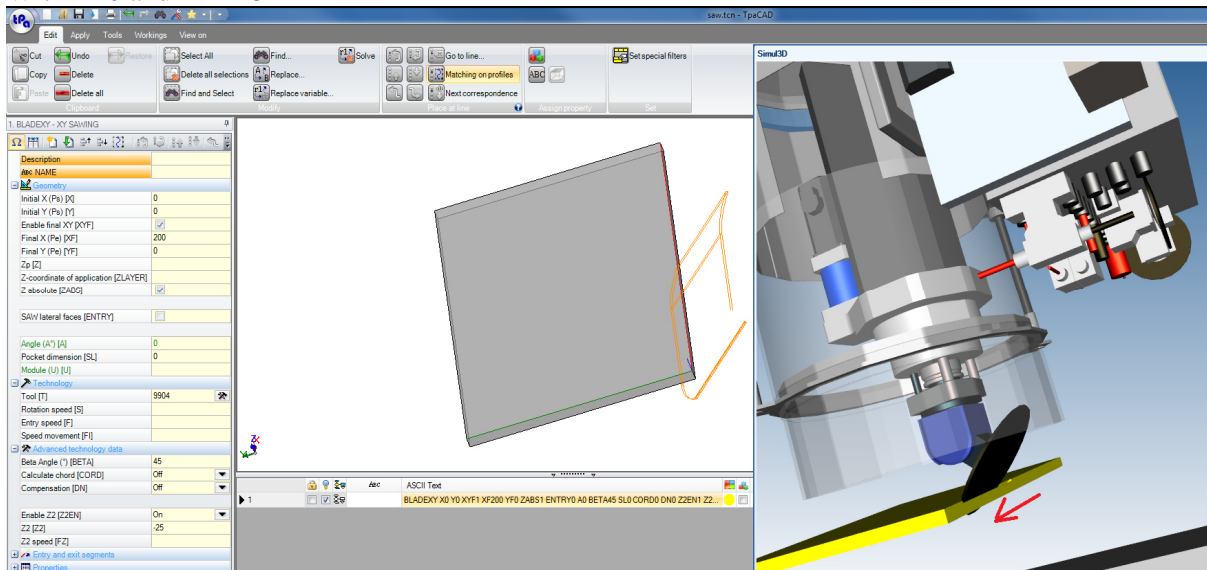
Here an example

Tool setting: saw rotation direction: **CLOCKWISE**:

With $Z=-25$

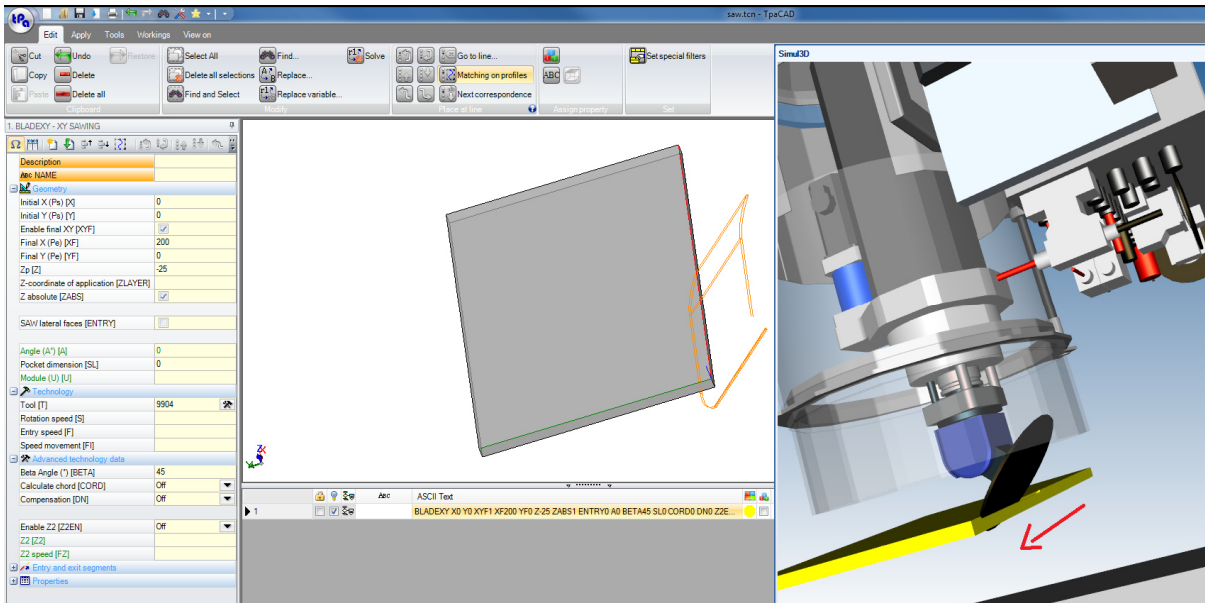


With $Z=0$ and $Z2=-25$

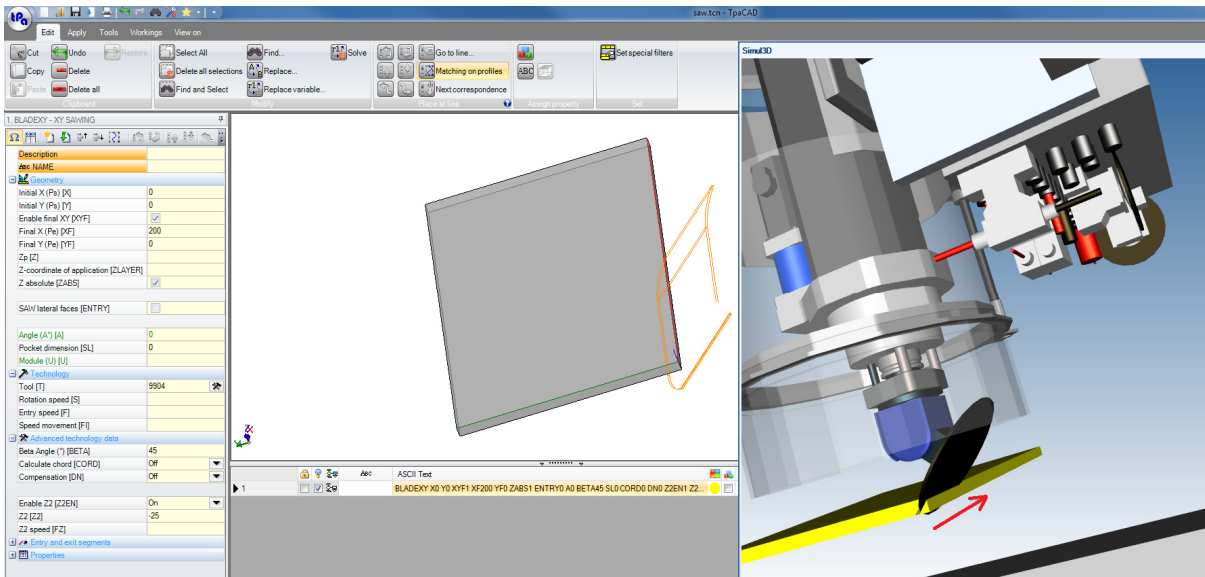


Tool setting: saw rotation direction COUNTERCLOCKWISE

Z=-25



With Z=0 and Z2=-25



Saw correction:

Left cut: without radius correction

Right cut: with radius correction

The correction is calculated axial to the sawblade => depth changes

